

Project 6: Book Connection Data Enrichment

Technical Specification Document

Organization: BIBLIOnomics

For: Dalhousie University

1 Executive Summary

1.1 The Business

BIBLIOnomics is a B2B book gifting platform. We help businesses give meaningful book gifts by matching books to recipients based on personal connections—shared geography, industry, interests, and more.

1.2 The Problem

Standard book metadata tells you *what* a book is. It doesn't tell you *who* it connects to.

1.3 The Solution

Valuable information is already in the metadata—we just need to extract and map it.

Book metadata contains rich, unstructured data about authors, locations, industries, and themes. This project extracts and structures that hidden value into connection data that powers our matching engine.

1.4 Project Scope

Build an enrichment system that:

- Fetches book metadata from BiblioShare API (primary source) using ISBNs
 - Supplements with Google Books and Open Library APIs (external sources)
 - Extracts and maps connection data (geography, industry, author info)
 - Outputs enriched data as a 75-column CSV file
- You deliver:** Source Code + Enriched CSV (500 books) + Connection Report

2 Input Specification

2.1 What You Receive

1. **ISBN List** (CSV file)—500 ISBNs to process, one per row
2. **BiblioShare API Token**—Authentication for BiblioShare API
3. **This Specification**—Defines 75 output fields and expected format

2.2 BiblioShare API (Primary Source)

Endpoint:

```
GET https://biblioshare.ca/BNCServices/BNCServices.aspx/ONIX
```

Parameters:

- **Token:** API authentication token (provided separately)

- EAN: ISBN-13 of the book to look up

Example Request:

```
https://bibliosshare.ca/BNCServices/BNCServices.asmx/ONIX?Token=[TOKEN]&
EAN=9780785291909
```

2.3 Example API Response

The BiblioShare API returns ONIX-formatted data. Below is a real example response (JSON format):

```
{
  "data": {
    "Product": {
      "RecordReference": [9780785291909],
      "ProductIdentifier": [
        {"ProductIDType": ["02"], "IDValue": ["0785291903"]},
        {"ProductIDType": ["03"], "IDValue": [9780785291909]}
      ],
      "ProductForm": ["BB"],
      "ProductFormDescription": ["Hardcover, Jacketed"],
      "Title": [{
        "TitleType": ["01"],
        "TitleText": ["Power Moves"],
        "Subtitle": ["Ignite Your Confidence and Become a Force"]
      }],
      "Contributor": [{
        "SequenceNumber": [1],
        "ContributorRole": ["A01"],
        "PersonName": ["Sarah Jakes Roberts"],
        "BiographicalNote": ["<p>Sarah Jakes Roberts is a <em>New York Times</em> bestselling author, speaker, entrepreneur, and philanthropist. She is the founder of Woman Evolve... Alongside her husband, Toure Roberts, she serves as an Assistant Pastor at The Potter's House Dallas...</p>"]
      }],
      "Language": [{"LanguageRole": ["01"], "LanguageCode": ["eng"]}],
      "NumberOfPages": [224],
      "BASICMainSubject": ["REL012130"],
      "Subject": [
        {"SubjectSchemeIdentifier": [10], "SubjectCode": ["REL050000"], "SubjectHeadingText": ["RELIGION / Christian Ministry / Counseling & Recovery"]},
        {"SubjectSchemeIdentifier": [10], "SubjectCode": ["SEL031000"], "SubjectHeadingText": ["SELF-HELP / Personal Growth / General"]}
      ],
      "OtherText": [
        {"TextTypeCode": ["01"], "Text": ["<p><strong>Unleash the superpower of being yourself...</strong></p>"]},
        {"TextTypeCode": ["02"], "Text": ["New full-length trade book from the bestselling author of Woman Evolve."]}
      ],
      "MediaFile": [{
        "MediaFileTypeCode": ["04"],
        "MediaFileLink": ["http://media.zondervan.com/.../9780785291909.jpg"]
      }],
    }
  },
}
```

```

    "Publisher": [{"PublisherName": ["Thomas Nelson"]}],
    "CityOfPublication": ["Nashville"],
    "CountryOfPublication": ["US"],
    "PublicationDate": [20240430],
    "Measure": [
      {"MeasureTypeCode": ["01"], "Measurement": [9.3],
       "MeasureUnitCode": ["in"]},
      {"MeasureTypeCode": ["02"], "Measurement": [6.3],
       "MeasureUnitCode": ["in"]},
      {"MeasureTypeCode": ["03"], "Measurement": ["0.90"],
       "MeasureUnitCode": ["in"]},
      {"MeasureTypeCode": ["08"], "Measurement": ["0.770"],
       "MeasureUnitCode": ["lb"]}
    ],
    "SupplyDetail": [{
      "Price": [
        {"PriceAmount": [36.99], "CurrencyCode": ["CAD"]},
        {"PriceAmount": [29.99], "CurrencyCode": ["USD"]}
      ]
    }]
  }
}

```

2.4 External Sources (Supplementary)

Google Books API

Endpoint: GET <https://www.googleapis.com/books/v1/volumes?q=isbn:{ISBN}>

No authentication required. Use for: averageRating, ratingsCount

Open Library API

Endpoint: GET <https://openlibrary.org/isbn/{ISBN}.json>

No authentication required. Use for: Availability check

Data Source Priority:

1. BiblioShare (Primary)—Canadian metadata, pricing, physical specs
2. Google Books (Secondary)—Ratings
3. Open Library (Tertiary)—Availability check

3 Complete Flow Example

This example shows the complete extraction and mapping flow for one book.

3.1 Step 1: Fetch Raw Data

Input ISBN: 9780785291909

Call BiblioShare API → Receive ONIX data (as shown in Section 2.3)

3.2 Step 2: Extract Source Fields (Section A)

From the ONIX response, extract and map:

title:	"Power Moves"
subtitle:	"Ignite Your Confidence and Become a Force"
primary_author:	"Sarah Jakes Roberts"
isbn_13:	"9780785291909"
isbn_10:	"0785291903"

```

book_format:      "Hardcover, Jacketed"
publisher:        "Thomas Nelson"
publication_date:  "2024-04-30"
page_count:       224
list_price_cad:   36.99
list_price_usd:   29.99

author_description (after HTML stripping):
"Sarah Jakes Roberts is a New York Times bestselling author,
speaker, entrepreneur, and philanthropist. She is the founder
of Woman Evolve... Alongside her husband, Toure Roberts, she
serves as an Assistant Pastor at The Potter's House Dallas..."

bisac_primary_code: "REL012130"
bisac_level_1:      "RELIGION"
bisac_level_2:      "Christian Ministry"
bisac_level_3:      "Counseling & Recovery"

```

3.3 Step 3: Extract Physical Dimensions (Section B)

From ONIX Measure array:

```

book_height_in:      9.3
book_width_in:       6.3
spine_thickness_in:  0.90
book_weight_lb:      0.770
cover_dimensions:    "9.3 x 6.3"
book_size_category:  "Standard" (Height 7-10" AND Width 5-7")

```

3.4 Step 4: Extract Author Bio Data (Section C)

Parse author_description using pattern matching:

```

Input text: "Sarah Jakes Roberts is a New York Times bestselling
author, speaker, entrepreneur, and philanthropist... serves as
an Assistant Pastor at The Potter's House Dallas..."

Extraction results:
author_location_raw:  "Dallas" (from "The Potter's House Dallas")
author_province:     "USA" (Dallas is not Canadian)
author_region:       "USA"
author_profession_raw: "bestselling author, speaker, entrepreneur,
and philanthropist"
author_institution:   "The Potter's House Dallas"
author_bio_richness:  "Rich" (word count > 100)

author_summary:      "Sarah Jakes Roberts is a USA-based author,
speaker, entrepreneur, and philanthropist
at The Potter's House Dallas"

```

3.5 Step 5: Derive Connection Fields (Section D)

Apply deterministic rules:

```

primary_industry:      "General Interest" (RELIGION -> General
Interest)
gift_price_tier:       "Premium" ($36.99 CAD in $35-50 range)

```

```

book_length_category:  "Standard" (224 pages in 150-300 range)
publisher_region:      "USA" (Thomas Nelson -> USA)
book_themes_bisac:     "RELIGION / Christian Ministry / Counseling"
locations_mentioned:   "Dallas" (from description)
time_period_mentioned: "" (no year/decade found)
corporate_appropriate: TRUE (no flagged BISAC codes)

```

3.6 Step 6: Fetch External Data (Section E)

```

Google Books API call -> Found
google_books_rating:    4.5
google_books_rating_count: 1247
google_books_available: TRUE

Open Library API call -> Found
open_library_available: TRUE

Constructed URLs:
amazon_ca_url:  "https://amazon.ca/dp/0785291903"
goodreads_url:  "https://goodreads.com/book/isbn/9780785291909"

```

3.7 Step 7: Generate Metadata (Section F)

```

enrichment_status:  "Complete"
enrichment_date:    "2026-01-21"
confidence_score:   85 (rich bio + external data available)
data_richness:      "Rich"
data_sources:       "BiblioShare|Google Books|Open Library"
fields_populated:   68
verification_status: "Verified"
last_updated:       "2026-01-21"

```

3.8 Step 8: Capture Extras (Section G)

```

extracted_extras:
"[UNVERIFIED] Other works: Woman Evolve
[UNVERIFIED] Role: founder of Woman Evolve
[UNVERIFIED] Spouse: Toure Roberts"

```

4 Field Definitions (75 Fields)

4.1 Section A: Source Data (Fields 1–35)

Passthrough from BiblioShare with minimal transformation.

1. **id**—Generate unique ID (UUID or sequential)
2. **title**—Book title
3. **primary_author**—Main author name
4. **isbn_13**—13-digit ISBN
5. **book_format**—Physical format (e.g., “Hardcover, Jacketed”)
6. **cdn_creator**—Boolean: TRUE if Canadian creator (from ONIX)
7. **short_description**—Marketing blurb (OtherText type 02)
8. **genre_tags**—Genre classification (from BISAC level 1)

9. `cover_source`—Cover image URL
10. `isbn_10`—10-digit ISBN
11. `subtitle`—Book subtitle
12. `series_title`—Series name if applicable
13. `publisher`—Publisher name
14. `author_description`—Raw author bio (strip HTML tags)
15. `awards`—Awards won
16. `bisac_primary_code`—Primary BISAC code (e.g., “REL012130”)
17. `bisac_secondary_code`—Secondary BISAC code
18. `bisac_tertiary_code`—Tertiary BISAC code
19. `bisac_level_1`—BISAC top category (e.g., “RELIGION”)
20. `bisac_level_2`—BISAC second level
21. `bisac_level_3`—BISAC third level
22. `bisac_level_4`—BISAC fourth level
23. `series_number`—Position in series
24. `volume_number`—Volume number
25. `table_of_contents`—TOC if available
26. `all_authors`—Comma-separated list of all authors
27. `secondary_author`—Second contributor
28. `publication_date`—Format: YYYY-MM-DD
29. `format`—Hardcover, Paperback, etc.
30. `long_description`—Full marketing description (strip HTML)
31. `page_count`—Number of pages
32. `target_audience`—Intended audience
33. `list_price_cad`—Canadian retail price
34. `list_price_usd`—US retail price
35. `reading_level`—Reading level if specified

BISAC Code Parsing: Use `SubjectHeadingText` field (e.g., “RELIGION / Christian Ministry / Counseling”) and split by “ / ”.

Resource: <https://www.bisg.org/complete-bisac-subject-headings-list>

HTML Stripping: ONIX text fields contain HTML tags. Strip all HTML before storing.

4.2 Section B: Physical Dimensions (Fields 36–45)

Extracted from ONIX Measure data.

36. `book_height_in`—Height in inches
37. `book_height_mm`—Height in millimeters
38. `book_width_in`—Width in inches
39. `book_width_mm`—Width in millimeters
40. `spine_thickness_in`—Spine in inches
41. `spine_thickness_mm`—Spine in millimeters
42. `book_weight_lb`—Weight in pounds
43. `book_weight_g`—Weight in grams
44. `cover_dimensions`—Format: “H x W in” (e.g., “9.5 x 6.3”)
45. `book_size_category`—Pocket / Standard / Large / Coffee Table / Unknown

ONIX Measure Type Codes: 01=Height, 02=Width, 03=Spine, 08=Weight

Book Size Category Logic:

- Pocket: Height < 7” OR Width < 5”
- Standard: Height 7–10” AND Width 5–7”
- Large: Height 10–12” OR Width 7–9”
- Coffee Table: Height > 12” OR Width > 9”
- Unknown: No dimension data available

4.3 Section C: Author Bio Extraction (Fields 46–52)

Extracted from `author_description` field using pattern matching.

46. `author_location_raw`—Location phrase verbatim (e.g., “from Halifax, NS”)
47. `author_province`—Province mapped from city (e.g., “Nova Scotia”)
48. `author_region`—Region derived from province (e.g., “Atlantic”)
49. `author_profession_raw`—Profession verbatim (e.g., “food writer”)
50. `author_institution`—Institution/employer if mentioned
51. `author_bio_richness`—Rich (>100 words) / Moderate (50–100) / Sparse (<50) / None
52. `author_summary`—Human-readable combined summary of author (see below)

Province Values: Nova Scotia, New Brunswick, Prince Edward Island, Newfoundland and Labrador, Quebec, Ontario, Manitoba, Saskatchewan, Alberta, British Columbia, Yukon, Northwest Territories, Nunavut, USA, International, Unknown

Region Mapping:

- Atlantic: NS, NB, PE, NL
- Quebec: QC
- Ontario: ON
- Prairies: MB, SK, AB
- British Columbia: BC
- North: YT, NT, NU

Author Summary Field:

Combine verified extraction data into a human-readable sentence:

Template: "[Name] is a [region]-based [profession] at [institution]"

Examples:

- "Elisabeth Bailey is an Atlantic-based food and gardening writer"
- "Dan Martell is a Canadian entrepreneur"
- "Sarah Jakes Roberts is a USA-based author, speaker, and philanthropist at The Potter's House Dallas"

Rules:

- Only include verified data from fields 46–51
- If location unknown, omit region
- If no institution, omit "at [institution]"
- If no profession, use "author"

Extraction Principle: If uncertain, leave blank. Never guess.

4.4 Section D: Connection Fields (Fields 53–60)

Deterministic rules and verifiable extraction.

53. `primary_industry`—Mapped from BISAC (see Appendix A)
54. `gift_price_tier`—Budget (<\$20) / Standard (\$20–35) / Premium (\$35–50) / Luxury (\$50+)
55. `book_length_category`—Quick Read (<150) / Standard (150–300) / Substantial (300–450) / Epic (450+)
56. `publisher_region`—Publisher mapped to region (see Appendix B)
57. `book_themes_bisac`—BISAC hierarchy concatenated
58. `locations_mentioned`—Places found in description, pipe-delimited
59. `time_period_mentioned`—Years/decades found in description
60. `corporate_appropriate`—Boolean: FALSE if BISAC contains flagged categories

4.5 Section E: External Sources (Fields 61–66)

From Google Books and Open Library APIs.

61. `google_books_rating`—Average rating from Google Books
62. `google_books_rating_count`—Number of ratings
63. `google_books_available`—Boolean: TRUE if found
64. `open_library_available`—Boolean: TRUE if found
65. `amazon_ca_url`—Constructed: `https://amazon.ca/dp/{ISBN-10}`
66. `goodreads_url`—Constructed: `https://goodreads.com/book/isbn/{ISBN-13}`

4.6 Section F: Enrichment Metadata (Fields 67–74)

System-generated fields.

67. `enrichment_status`—Complete / Partial / Failed / Pending
68. `enrichment_date`—YYYY-MM-DD
69. `confidence_score`—0–100 calculated score
70. `data_richness`—Rich / Moderate / Sparse / Minimal
71. `data_sources`—Pipe-delimited: BiblioShare—Google Books—Open Library
72. `fields_populated`—Count of non-empty fields (0–75)
73. `verification_status`—Verified / Partial / Unverified
74. `last_updated`—YYYY-MM-DD

4.7 Section G: Extracted Extras (Field 75)

Catch-all for unverified data.

75. `extracted_extras`—Unstructured data with [UNVERIFIED] labels

5 Deliverables

5.1 Primary Deliverables

1. Source Code

- Complete, runnable enrichment system
- README with setup instructions
- Runnable with single command (e.g., `python enrich.py input.csv output.csv`)
- Include lookup tables you created (BISAC mapping, publisher regions, etc.)

2. Enriched CSV

- 500 books processed
- One row per ISBN, 75 columns as specified
- UTF-8 encoding

3. Connection & Relationship Report

- Demonstrate that your system extracts meaningful connection data
- Show geographic, industry, and Canadian content connections
- Include data quality metrics

5.2 Supporting Deliverables

4. Architecture Documentation (1–2 pages)
5. Setup & Run Guide
6. Live Demo

A BISAC to Industry Mapping (Sample)

- BUSINESS & ECONOMICS → Finance & Banking
- COMPUTERS → Technology
- COOKING → Agriculture & Food
- EDUCATION → Education
- HEALTH & FITNESS → Healthcare & Pharma
- LAW → Legal
- MEDICAL → Healthcare & Pharma
- POLITICAL SCIENCE → Government & Public Sector
- TECHNOLOGY & ENGINEERING → Technology
- TRAVEL → Tourism & Hospitality
- GARDENING → Agriculture & Food
- ARCHITECTURE → Real Estate
- HOUSE & HOME → Real Estate
- CRAFTS & HOBBIES → Retail & Consumer
- PETS → Retail & Consumer
- TRANSPORTATION → Manufacturing
- JUVENILE → Education
- All others → General Interest

B Publisher to Region Mapping (Sample)

Atlantic: Nimbus Publishing, Pottersfield Press, Goose Lane Editions, Breakwater Books, Flanker Press, Acorn Press, Cape Breton University Press, Formac Publishing

Quebec: Vehicule Press, McGill-Queen's University Press, Linda Leith Publishing, Baraka Books

Ontario: House of Anansi, ECW Press, Dundurn Press, Coach House Books, University of Toronto Press, Cormorant Books, Second Story Press, Random House Canada, Penguin Canada, HarperCollins Canada

Prairies: Thistledown Press, NeWest Press, Great Plains Publications, University of Alberta Press, University of Manitoba Press, Freehand Books

British Columbia: Talonbooks, Douglas & McIntyre, Harbour Publishing, Greystone Books, Arsenal Pulp Press, New Star Books, Orca Book Publishers

USA: Thomas Nelson, Penguin Publishing Group, HarperCollins, Simon & Schuster, Random House

Unknown publishers default to "Unknown" region.

C Province to Region Mapping

- Atlantic: Nova Scotia, New Brunswick, Prince Edward Island, Newfoundland and Labrador
- Quebec: Quebec
- Ontario: Ontario
- Prairies: Manitoba, Saskatchewan, Alberta
- British Columbia: British Columbia
- North: Yukon, Northwest Territories, Nunavut
- USA: United States locations
- International: All other countries

D Common Extraction Patterns

Location in Author Bio:

- “from [City], [Province]”
- “based in [City]”
- “lives in [City]”
- “[City] native”
- “at [Organization] in [City]”

Profession in Author Bio:

- “[Name] is a [profession]”
- “[Name], a [profession],”
- “award-winning [profession]”
- “[profession] and author”

Institution in Author Bio:

- “at [Organization]”
- “founder of [Organization]”
- “serves as [Role] at [Organization]”
- “teaches at [University]”

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